MD FAJLA RABBI

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LinkedIn | Google Scholar

EDUCATION

PhD in Mechanical Engineering, Clemson University (Fall 2026-present)

M.Sc. in Mechanical Engineering, University of Ulsan (2023-2025)

B.Sc. in Mechanical Engineering, Khulna University of Engineering & Technology (2015-2019)

RESEARCH EXPERIENCE

Graduate Research Assistant

FPMI lab, Department of Mechanical Engineering, University of Ulsan

Advisor: Prof. Kyoung Kwan Ahn (Google scholar)

- Fabricated electrospun TiO2-PVDF nanocomposite membranes to optimize β-phase content, surface roughness, and surface charge density for liquid-solid triboelectric nanogenerator (TENG) devices.
- Characterized β-phase content using XRD (Rigaku Ultima IV) and FTIR (Thermo Fisher Nicolet iS5), morphology using FE-SEM (JEOL JSM-7600) with EDS mapping, and surface roughness using AFM (MFP-3D).
- Measured membrane hydrophobicity by water contact angle using a SmartDrop analyzer (FemtoFAB), and dielectric constant using an Agilent 4294A precision impedance analyzer.
- Built droplet-based liquid-solid triboelectric nanogenerator (TENG) devices and designed a 3D-printed cascaded structure with two stacked membranes.
- Measured open-circuit voltage (Voc), short-circuit current (Isc), and transferred charge using a Keithley DMM7510 digital multimeter.
- Wrote a first-author review on electrospun nanofibers for TENGs (Advanced Materials, 2026).

JOURNAL AND CONFERENCE PROCEEDINGS

- **Rabbi, MD Fajla**, Ahn JH, Duy Linh Vu, and Kyoung Kwan Ahn. "Recent Advances in Electrospun Nanofibers for Triboelectric Nanogenerators: Performance Enhancement Strategies and Emerging Applications". *Advanced materials* (2026). https://doi.org/10.1002/adma.202522834
- **Rabbi, MD Fajla,** Duy Linh Vu, and Kyoung Kwan Ahn. "Electrospun TiO2–PVDF Nanocomposite Membranes for High-Performance Liquid–Solid Triboelectric Nanogenerators with Cascaded Architecture." *Journal of Alloys and Compounds* (2025). https://doi.org/10.1016/j.jallcom.2025.182531
- Ahn JH, Nguyen QT, Nguyen TBT, **Rabbi MF**, Nguyen VH, Lee YH, Ahn KK. "Direct vs. Indirect Charge Transfer: A Paradigm Shift in Phase-Spanning Triboelectric Nanogenerators Focused on Liquid and Gas Interfaces." *Energies* (2025). https://doi.org/10.3390/en18215709
- **Rabbi, MD Fajla**, Kyoung Kwan Ahn. "Self-Powered Acid Rain Sensor Based on Triboelectric Nanogenerator Made of Electrospun TiO_2-PVDF Composite film" 27th International Conference on Mechatronics Technology (ICMT), 2024(Kanazawa, Japan).

AWARDS & FELLOWSHIPS

BK21 Scholarship	Government of Korea	2023 - 2025
Munsu AF-1 Scholarship	University of Ulsan	2023 - 2024
Department Merit Scholarship	KUET	2017 - 2018

TECHNICAL SKILLS

Category	Methods, tools, and applications
Fabrication	Electrospinning (Inovenso NanoSpinner NS+), blade coating
Electrical characterization	Open-circuit voltage (Voc), short-circuit current (Isc), and charge transfer of TENGs using a Keithley DMM7510; frequency-dependent dielectric constant (Agilent 4294A impedance analyzer)
Materials characterization	FE-SEM (JEOL JSM-7600), AFM (MFP-3D), XRD (Rigaku Ultima IV), FTIR (Thermo Fisher Nicolet iS5), water contact angle (FemtoFAB SmartDrop), and optical microscopy
Simulation	COMSOL Multiphysics (TENG electrostatic modeling); ANSYS Fluent (thermal analysis of PCM-based heat sinks)
Programming	Python (PEMFC stack model: activation, ohmic, and concentration polarization losses); MATLAB
Design and prototyping	SolidWorks, Autodesk Inventor, and 3D printing

TEACHING SKILLS

Graduate Laboratory Assistant, ME 4440: Mechanical Engineering Laboratory III, Clemson University (Fall 2026-present)
Lab topics: vibration modal analysis, mass-pendulum system, PLC security system, convective heat transfer, and fatigue testing.

PROFESSIONAL EXPERIENCE

Project Engineer (MPVC Limited / licensor Oxy Vinyls LP, Texas) 2019-2023

- Worked on installation, commissioning, and mechanical completion of a 400 TPD vinyl chloride monomer (VCM) plant, including an EDC cracking furnace (Heurtey Petrochem) and an incineration unit (John Zink).
- Installed and commissioned a 20,000 m^3 cryogenic ethylene storage tank with three BOG compressors and a propylene refrigeration unit, and a 100 TPD cryogenic air separation unit (Linde).
- Handled equipment and piping in SS 304, 310, and 316L, Incoloy, duplex stainless steel, and Monel; checked material certificates and traceability.
- Coordinated radiographic (RT) and dye penetrant (PT) testing of welds and tracked repairs and re-examination.
- Basic working knowledge of ASTM material standards and ASME/API codes for pressure vessels and storage tanks.
- Took part in pre-commissioning of piping and equipment, including line flushing, leak testing, and no-load run tests of pumps and compressors.
- Supported plant startup by checking equipment during the first operating runs and helping fix mechanical issues such as leaks, vibration, and seal failures.